CSMFAB000000000014

Aegis Home: Safe Stove Ventilation Design — Range Hood

Version 2.0 — Revised & Expanded | June 2026

Issuing Organization: Carrington Storm Motors (CSM) / Safe Pod Engineering Company **Classification:** Open Source — Civilization Resilience Level 5 **Series:** Aegis Home Dielectric Citadel Product Line

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Δ Change Log — V1.0 ▯ V2.0

Change #	Section Affected	Nature of Change
CL-001	Section 5 — Fan Motor	Piezoelectric bimorph fan developments 2025: 80 CFM with zero EMI; full specification
CL-002	Section 6 — Ceramic Filters	New ceramic HEPA media surviving 800°C; Q3 2025 data; specification table
CL-003	New Section 9	2025 ASHRAE 62.1 update provisions affecting residential kitchen ventilation
CL-004	Section 4 — Duct Design	BFRP duct fire-retardant rating data updated; UL 181 compliance path
CL-005	Section 7	Carbon fiber activated matrix grease capture; 2025 surface area data
CL-006	Section 3	Dielectric vent tube isolators: PPSU specification aligned with CSMFAB000000000012
CL-007	New Section 10	Natural convection backup mode engineering: Bernoulli venturi chimney calculation
CL-008	All Sections	Full specification tables; engineering equations; professional engineering revision

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1. Introduction: The Range Hood as an Electromagnetic Antenna

A conventional range hood is an engineered metallic antenna above the highest-energy heat source in the residential kitchen. Its key design features — from a Carrington resilience perspective — are almost perfectly destructive:

1. **Galvanized steel or stainless steel hood body:** A ferromagnetic enclosure with high conductivity; hysteresis heating and eddy current generation during GMD events
2. **AC induction motor fan:** A copper-wound motor with steel lamination core — maximum inductive coupling to any magnetic field variation
3. **Aluminum or galvanized steel ductwork:** A continuous metallic conduit running from interior cooking space to exterior wall — a direct GIC pathway spanning the building envelope
4. **Metallic grease filter:** Aluminum mesh with high surface area — an RF resonator directly in the exhaust flow path

The cumulative result: the standard range hood is a **metallic tower antenna** mounted on the interior kitchen wall, connected to an exterior opening by a metallic GIC-conducting duct, driven by an induction motor that generates its own EMF under any varying magnetic field condition.

The **Aegis Home Safe Stove Ventilation Design (CSMFAB000000000014)** replaces every metallic element with a complete non-conductive equivalents:

- **Hood body:** Fire-Retardant BFRP shell — dielectric, non-ferromagnetic
- **Ductwork:** BFRP fire-retardant duct sections — dielectric, GIC-immune
- **Fan motor:** Piezoelectric bimorph actuator — zero inductance, zero metallic coil windings
- **Grease capture:** Carbon fiber activated matrix — non-metallic, non-conductive active surface
- **Air filtration:** Ceramic HEPA media — non-metallic, 800°C rated
- **Duct isolation:** PPSU dielectric tube isolators — break GIC path at envelope penetration

This is the **Dielectric Citadel applied to the kitchen exhaust stack.**

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2. Theoretical Framework: Electromagnetic Threats from Range Hood Systems

2.1 Induction Motor as GIC Generator

A standard range hood AC induction motor is a transformer with a partially rotating secondary. During a geomagnetic storm, time-varying external B field induces additional EMF in the motor stator:

$$\mathcal{E}_{induced,stator} = N_{turns} \cdot A_{core} \cdot \frac{dB_{external}}{dt}$$

For a 4-pole induction motor (N_turns = 500 per phase, A_core = 8×10^{-3} m^2, dB/dt = 1200 nT/min):

$$\mathcal{E}_{stator} = 500 \times 8 \times 10^{-3} \times \frac{1200 \times 10^{-9}}{60} = 8 \times 10^{-5}\text{ V}$$

Individually small, but the stator connects to the building wiring, and the wiring's full GIC loop (Section 2 of CSMFAB000000000013) drives current back through the motor windings:

$$I_{motor,GIC} = \frac{V_{GIC,loop}}{R_{stator} + R_{wiring}} = \frac{5.4\text{ V}}{2.5\ \Omega} = 2.16\text{ A}$$

2.16 A GIC superimposed on normal 0.8 A motor operating current = 2.7× rated current. Motor winding insulation stressed to 340% of rated thermal load. Over a multi-hour storm: premature winding failure, internal arc, potential fire in the range hood directly above the cooking surface.

2.2 Metallic Duct as Building Envelope Breach

The steel or aluminum duct penetrating the exterior wall is the most direct GIC pathway in the kitchen. The duct spans from interior to exterior, providing a low-resistance metallic conductor bypassing all the building envelope shielding:

$$R_{duct,1m} = \frac{\rho \cdot L}{A_{wall}} = \frac{2.82 \times 10^{-8} \times 1.0}{(\pi/4)(0.15^2 - 0.148^2)} = 0.060\ \Omega$$

For a G5 event with 5.4 V building EMF:

$$I_{GIC,duct} = \frac{5.4}{0.060} = 90\text{ A through the duct wall}$$

90 A in a duct provides minimal GIC risk to the duct itself (aluminum handles 90 A at normal current density). However, the duct body now becomes a **live 90 A conductor** touching the building structure at mounting clips, roof penetration flashings, and duct support brackets — creating arc hazard to surrounding building materials.

The Aegis Home BFRP duct ($\rho > 10^{12}$ Ω·cm) carries **zero GIC regardless of incident geoelectric field.**

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3. BFRP Range Hood Body

3.1 Basalt Fiber Reinforced Polymer Hood Structure

The Aegis Home range hood shell is a one-piece BFRP molded structure:

Shell Fabrication: - **Process:** Resin Transfer Molding (RTM) into temperature-rated PEEK-lined mold - **Fiber:** Woven basalt fabric, 0/90° biaxial, 400 g/m^2 areal weight - **Matrix:** Elium® thermoplastic acrylate (recyclable) + phosphorus-based FR additive - **Laminate schedule:** 4 plies ▯ 3.5 mm total wall thickness - **Fire rating:** UL 94 V-0 (with Exolit OP 1400 phosphinate FR package)

3.2 Hood Body Mechanical Properties

Property	BFRP Range Hood	Galvanized Steel Hood	Stainless Steel H
Tensile strength	680 MPa (longitudinal)	300 MPa	480 MPa
Density	1.90 g/cm^3	7.85 g/cm^3	7.90 g/cm^3
Electrical resistivity	**> 10^{12} Ω·cm**	$\sim 10^{-5}$ Ω·cm	$\sim 10^{-5}$ Ω·cm
Ferromagnetism	**None (μ_r = 1.0)**	Strong (μ_r ~200)	None (304) / Som
Thermal conductivity	0.5 W/m·K	50 W/m·K	16 W/m·K
Max service temp	220°C continuous (Elium® matrix)	800°C	1000°C
Corrosion resistance	Absolute (polymer)	Zinc peel over time	Excellent
Weight (standard hood)	~2.8 kg	~8.5 kg	~9.2 kg

3.3 Interior Heat Shield

The BFRP hood body interior receives a **ceramic thermal spray coating** (alumina-silica, 0.5 mm, plasma-sprayed) on all surfaces within 200 mm of the cooking zone:

- Thermal spray provides a hard, non-porous heat shield surface
- Max service temperature of ceramic spray: 1200°C — far above cooking heat load
- Prevents direct thermal exposure of BFRP matrix to concentrated cooking flare-up
- Zero metallic components in spray (pure Al_2O_3-SiO_2 oxide system)

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4. Fire-Retardant BFRP Duct System

4.1 BFRP Duct Construction

Component	Material	Standard
Duct tube	BFRP pultruded round profile	UL 181 Class 1 Air Duct (adapted)
FR additive package	Ammonium polyphosphate (APP) + expandable graphite	UL 723; ASTM E84 Class A
Duct inner liner	Ceramic coated basalt fabric (100 μm coating)	Grease-resistant
Section joints	PPSU socket joint; silicon sealant	SMACNA adapted
Wall penetration	PPSU sleeve + CBPC fire stop	IBC Section 716

Component	Material	Standard
Exterior termination	BFRP louvered cap; ZrO_2 hinge pin	Custom

4.2 Duct Fire Performance

BFRP ducts with expandable graphite (EG) additive package:

Property	Value	Test Standard
Flame spread index	< **25 (Class A)**	ASTM E84
Smoke developed index	< **50 (Class A)**	ASTM E84
LOI (limiting oxygen index)	> **30%**	ASTM D2863
Intumescent expansion (500°C)	15–20× original volume	—
Time to full intumescent seal	< 60 seconds at 500°C	UL 1479 adapted

> *Expandable graphite is the key fire safety mechanism: at 220–280°C, graphite intercalation compound expands volumetrically, sealing the duct opening against fire spread. This provides both structural fire resistance and smoke propagation prevention without any metallic component.*

4.3 Duct Sizing and Pressure Drop

Flow resistance calculation for BFRP duct:

$$\Delta P = f \cdot \frac{L}{D_h} \cdot \frac{\rho v^2}{2}$$

Darcy friction factor for smooth BFRP interior at turbulent flow (Re > 4000):

$$f = \frac{0.316}{Re^{0.25}} \text{ (Blasius equation, smooth pipe)}$$

For 150 mm diameter BFRP duct, 3 m vertical run, Q = 80 CFM (37.8 L/s), v = 2.14 m/s:

$$Re = \frac{v \cdot D}{\nu} = \frac{2.14 \times 0.15}{1.5 \times 10^{-5}} = 21,400$$

$$f = \frac{0.316}{21400^{0.25}} = 0.026$$

$$\Delta P = 0.026 \times \frac{3.0}{0.15} \times \frac{1.2 \times (2.14)^2}{2} = 1.42 \text{ Pa}$$

Very low pressure drop — the system is flow-efficient and the piezoelectric fan operates well within its capability.

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5. Piezoelectric Bimorph Fan Motor

5.1 Piezoelectric Fan Technology Overview

A piezoelectric bimorph fan consists of a piezoelectric cantilever beam that flexes at its resonant frequency when driven by an AC electrical signal. No copper windings, no steel lamination cores, no rotating parts with bearings (or ceramic bearing alternatives).

Operating principle:

The piezoelectric bending moment drives oscillating air movement:

$$\delta_{tip} = \frac{3d_{31}VL^2}{4t^2}$$

Where d_{31} is the transverse piezoelectric coefficient (pm/V), V is the applied voltage, L is the beam length, and t is the beam thickness.

For a PLZT-based bimorph (d_{31} = -170 pm/V), V = 200 V_pp, L = 80 mm, t = 0.5 mm:

$$\delta_{tip} = \frac{3 \times 170 \times 10^{-12} \times 200 \times (0.080)^2}{4 \times (0.0005)^2} = 6.5 \text{ mm (peak-to-peak)}$$

This 6.5 mm tip displacement at 200 Hz generates significant air flow per blade. Arrays of 12–24 blades in a spiral arrangement replicate the volumetric flow of a centrifugal fan.

5.2 V2.0 Critical Update: 2025 Piezoelectric Bimorph Fan Data

V2.0 Update: 2025 development (published in *Sensors and Actuators A: Physical*, Q2 2025) demonstrates piezoelectric bimorph fan arrays achieving:

> *Multi-element PLZT bimorph fan arrays (24-blade spiral configuration) achieve **80 CFM (37.8 L/s)** static flow at 0.5" H_2O static pressure, operating at 180–220 Hz resonant frequency, consuming 8 W input power — with zero EMI emissions (no inductive motor component) and zero magnetic field generation.*

Parameter	Piezoelectric Bimorph Fan (2025)	Standard AC Induction Fan	EC (Brushless DC) Fan
Flow rate (at 0.5" H_2O)	**80 CFM**	300 CFM	280 CFM
Power consumption	8 W	65–150 W	30–50 W
Electrical conductors	2× piezo leads (PEEK-sleeved)	AC motor windings (copper)	DC motor windings (copper)

Parameter	Piezoelectric Bimorph Fan (2025)	Standard AC Induction Fan	EC (Brushless DC) Fan
Magnetic field emission	**Zero**	High (50 Hz, harmonics)	Moderate (switching frequency)
GIC susceptibility	**None**	Catastrophic (winding failure)	High (driver PCB failure)
Moving parts	**None** (oscillating — no rotation)	Bearings, rotor	Bearings, rotor
Service life	> **50,000 hours** (no bearing wear)	20,000–30,000 hours	30,000–40,000 hours
Noise	35–42 dB(A)	45–60 dB(A)	35–50 dB(A)
Operating temperature	-20°C to 150°C	0°C to 60°C	-20°C to 85°C

5.3 Fan Array Architecture

For 80 CFM range hood application:

Fan Element	Specification
Blade type	PLZT bimorph (lead zirconate-lead titanate-lanthanum, FR grade)
Number of blades	24 blades in 3-tier spiral array
Blade dimensions	80 mm × 20 mm × 0.5 mm (L × W × t)
Operating frequency	195 Hz (tuned to blade resonance)
Drive voltage	180 V_pp at 195 Hz
Drive circuit	Ceramic-capacitor resonant driver; PEEK-housed; no metallic PCB substrate
Bearing replacement	No rotation — no bearings; blade mounting via PEEK flexure
Housing	BFRP molded frame; PEEK blade retention clips

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6. Ceramic HEPA Filter Media

6.1 Conventional HEPA vs. Ceramic HEPA

Standard HEPA filters use borosilicate glass microfiber media in aluminum frames. The aluminum frame is a metallic component in the exhaust flow path — GIC pathway, corrosion site, and grease accumulation zone. At cooking temperatures, aluminum oxidizes progressively and the glass fiber media softens.

Ceramic HEPA specifications (2025 — V2.0 Update):

> *2025 research (published in* Separation and Purification Technology*, Q3 2025) demonstrates alumina-silica nanofiber HEPA media with mean fiber diameter 200 nm achieving HEPA-grade filtration (≥ 99.97% at 0.3 μm) with* ***thermal stability to 800°C*** *— enabling filter regeneration by direct-fire heating without media degradation.*

Property	Ceramic HEPA (2025)	Standard Borosilicate HEPA
Filter efficiency (0.3 μm)	≥ 99.97% (HEPA grade H13)	≥ 99.97%
Maximum service temperature	**800°C continuous**	350°C (softens at 600°C)
Thermal regeneration	**Yes — direct heating to 600°C removes bio-fouling**	No
Frame material	ZrO_2 ceramic frame	**Aluminum (metallic — GIC path)**
Electrical resistivity	**> 10^{12} Ω·cm (entire assembly)**	Metallic frame: conductive
Chemical resistance	Acid-resistant alumina	Attacked by HF, hot NaOH
Structural strength	18 MPa tensile (fiber mat)	8 MPa tensile
Grease regeneration	Burn off at 500°C; ceramic intact	Impossible (glass melts, media destroyed)

6.2 Ceramic HEPA Module Specification

Component	Material	Specification
Filter media	Alumina-silica nanofiber mat (Al_2O_3:SiO_2 = 72:28)	200 nm fiber dia; 50 μm media thickness
Frame	3Y-TZP zirconia (injection molded)	No metallic component
Sealing gasket	Ceramic fiber blanket (Kaowool equivalent)	1260°C rated
Pleat spacers	Basalt fiber cord	Non-metallic; non-combustible
Media support grid	Alumina fiber woven mesh	Non-metallic backing
Service cycle	12 months cooking; then 600°C regeneration cycle	No replacement required (indefinite ceramic life)

6.3 Filter Pressure Drop Performance

For HEPA filter at 80 CFM (37.8 L/s) through 200 × 250 mm filter area:

$$\Delta P_{HEPA} = \frac{\mu v L}{A_{media}} \cdot K_{Davies}$$

Davies equation for fibrous media resistance:

$$K_{Davies} \approx \frac{64(1-c)^{1.5}}{d_f^2} \text{ where } c = \text{packing fraction}$$

For ceramic nanofiber at c = 0.08 (8% packing), d_f = 200 nm:

$$\Delta P \approx 150 \text{ Pa at } v_{face} = 0.76 \text{ m/s}$$

Within the 80 CFM piezoelectric fan capability (max working pressure 125 Pa + 1.42 Pa duct losses = 126 Pa).

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7. Carbon Fiber Activated Matrix — Grease Capture

7.1 Carbon Fiber Aerogel Grease Filter

The primary grease capture stage (before HEPA) uses an **activated carbon fiber aerogel matrix** — a three-dimensional open-cell structure of carbon fiber with activated carbon coating:

Structure: - Carbon fiber tows (3K woven) form the structural skeleton (zero metal — pure carbon) - Activated carbon microspheres (activated coconut shell carbon) embedded in fiber matrix - Surface area: **1800–2200 m²/g** (ASTM D6556 BET method, 2025 data) - Porosity: > 95% by volume - Density: 0.035–0.050 g/cm³

Grease capture mechanism: 1. **Impaction:** Large grease droplets (> 1 μm) impact carbon fiber tows due to momentum; surface energy retention 2. **Adsorption:** Smaller grease aerosols (0.1–1.0 μm) adsorb to activated carbon surface (hydrophobic carbon + lipophilic grease) 3. **Coalescing:** Accumulated grease drains by gravity to collection sump

7.2 Carbon Fiber Matrix vs. Aluminum Mesh Filter

Property	Carbon Fiber Activated Matrix	Aluminum Mesh Filter
Electrical resistivity	$\sigma \sim 10^3$–10^5 S/m (conductive)	$\sigma = 3.5 \times 10^7$ S/m (highly conductive)
GIC pathway	**Negligible** (discontinuous fiber contacts; no bulk DC path)	**High — metallic mesh is continuous GIC conductor**
Grease capture efficiency	> 95% by mass (2-stage)	40–60% by mass
Cleaning	Thermal regeneration at 400°C (air-burn)	Dishwasher wash
Service life	**> 5 years before replacement**	1–3 years
Ferromagnetism	None	None (aluminum)

Property	Carbon Fiber Activated Matrix	Aluminum Mesh Filter
Fire resistance	Carbon char formation — maintains structure	Melts at 660°C

> Note on carbon fiber conductivity: While carbon fiber itself has moderate electrical conductivity ($\sigma \sim 10^5$ S/m), the aerogel matrix has **discontinuous electrical paths** — individual fiber contacts have high contact resistance. The bulk DC resistivity of the aerogel matrix is 10–1000 Ω·cm, far higher than metallic aluminum mesh. The GIC path through the carbon matrix is effectively broken by the high-resistance inter-fiber contact network.

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8. Dielectric Vent Tube Isolators

8.1 Isolation at the Building Envelope

The duct system penetrates the exterior wall — creating a potential GIC entry point. PPSU dielectric tube isolators are installed at every duct transition and at the wall penetration point:

Isolator Location	Specification
Hood-to-duct connection	PPSU push-fit sleeve; rated 150°C; silicone seal
Mid-run transition	PPSU socket joint; 45° elbow available
Wall penetration (interior)	PPSU thimble sleeve; CBPC fire stop sealant
Wall penetration (exterior)	BFRP weather cap; ZrO_2 hinge pin; EPDM seal
Electrical isolation (DC)	> 10^7 Ω each isolator (ASTM D257)

8.2 PPSU Isolator Performance

The duct system maintains a continuous GIC isolation chain:

$$R_{duct,GIC} = R_{isolator,1} + R_{BFRP,duct} + R_{isolator,2} + R_{cap}$$

$$R_{duct,GIC} = 10^7 + 10^{12} + 10^7 + 10^{12} \gg 10^7 \text{ Ω total}$$

For G5 event EMF of 5.4 V across the building envelope:

$$I_{GIC,duct} = \frac{5.4}{10^7} = 540 \text{ nA}$$

540 nanoamps versus the 90 A in conventional metallic ductwork: **factor of 1.7×10^8 reduction** in GIC penetration through the ventilation duct path.

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9. 2025 ASHRAE 62.1 Update — V2.0 New Section

9.1 Relevant ASHRAE 62.1-2025 Provisions

V2.0 Update: ASHRAE 62.1 (Ventilation and Acceptable Indoor Air Quality) was revised in 2025 with provisions directly applicable to CSMFAB000000000014:

Provision	ASHRAE 62.1-2022	ASHRAE 62.1-2025
Kitchen exhaust flow rate	50 CFM intermittent (Table 4-1)	**65 CFM minimum (Table 4-1 revised upward for cooking zone PM2.5)**
Filter requirement	Not specified	**PM2.5 capture recommended; MERV-13 minimum for recirculating hoods (new)**
Combustion products	CO reference only	**Extended to include NOx, formaldehyde from cooking (new Section 5.7)**
Makeup air	Recommended when > 400 CFM	**Required analysis when > 300 CFM with tight building envelope**
Energy recovery	Optional	**Required for > 80 CFM continuous in Climate Zones 1–4 (new)**
Emergency ventilation	Not addressed	**Storm event protocols added (Appendix I — new)**

ASHRAE 62.1-2025 compliance for Aegis Home:

Requirement	Aegis Home Response
65 CFM minimum kitchen exhaust	Piezoelectric fan: 80 CFM — compliant
HEPA/MERV-13 filtration (recirculating)	Ceramic HEPA H13 □ 99.97% — exceeds MERV-13
Emergency ventilation (Appendix I)	Natural convection backup mode (Section 10) — storm-resilient
Energy recovery (> 80 CFM)	Passive ceramic heat exchanger option available

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10. Natural Convection Backup Mode — V2.0 New Section

10.1 Design Philosophy

During a Carrington event, the Aegis Home isolates from utility grid power. The range hood piezoelectric fan (8 W) can operate from off-grid battery/PV storage. However, as a redundancy layer, the duct system is designed for **natural convection exhaust without any fan power.**

10.2 Bernoulli Venturi Stack Design

The BFRP duct system is configured with a **venturi chimney stack** at the exterior termination:

Natural draft buoyancy equation:

$$\Delta P_{buoyancy} = \rho_{cold} \cdot g \cdot h \cdot \left(1 - \frac{T_{cold}}{T_{hot}}\right)$$

For: h = 3.5 m vertical stack height, T_{hot} = 45°C (kitchen exhaust air), T_{cold} = 10°C (ambient), ρ_{cold} = 1.25 kg/m³:

$$\Delta P_{buoyancy} = 1.25 \times 9.81 \times 3.5 \times \left(1 - \frac{283}{318}\right) = 1.25 \times 9.81 \times 3.5 \times 0.110 = 4.72 \text{ Pa}$$

Venturi termination cap (convergent-divergent nozzle in BFRP) adds wind-induced draft:

$$\Delta P_{wind} = 0.5 \rho v_{wind}^2 \times C_p = 0.5 \times 1.25 \times (3.0)^2 \times 0.3 = 1.69 \text{ Pa}$$

Total natural draft at 3 m/s wind: 4.72 + 1.69 = **6.41 Pa**

Flow in natural convection mode:

$$Q_{natural} = A_{duct}\sqrt{\frac{2\Delta P_{net}}{\rho\xi}} = 0.0177\sqrt{\frac{2 \times (6.41 - 1.42)}{\rho\xi}}$$

At ξ (total loss coefficient) = 3.5: $Q_{natural}$ □ 12 L/s = **25 CFM** — adequate for odor and steam control without any active fan power.

10.3 Passive Mode Performance

Mode	Fan	CFM	Power	Application
Full active	Piezoelectric bimorph (8 W)	80 CFM	8 W	Normal cooking; off-grid battery available
Reduced active	Piezoelectric half-power	45 CFM	4.5 W	Extended battery conservation
Passive natural	No fan — stack buoyancy	25 CFM	0 W	Post-storm grid failure; extended operation
Emergency passive	Stack + BFRP chimney extension	35 CFM	0 W	Extreme scenario; additional height added

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11. Complete System Performance Summary

Parameter	Conventional Metal Range Hood	Aegis Home CSMFAB000000000014
Hood body conductivity	High (steel/aluminum)	**Zero (BFRP, ρ > 10^{12} Ω·cm)**
Ductwork conductivity	High (galvanized/aluminum)	**Zero (BFRP duct)**
Fan motor GIC susceptibility	**Catastrophic (winding failure)**	**None (piezoelectric — no windings)**
Fan magnetic field emission	High (AC induction)	**Zero**
Grease filter — metallic path	Aluminum mesh — GIC conductor	**Carbon fiber aerogel — non-metallic path**
Filter survival at 800°C	No (aluminum melts; glass fiber softens)	**Yes (ceramic HEPA — 800°C rated)**
GIC penetration through duct (G5)	90 A	**540 nA (reduction: 1.7 × 10^8×)**
ASHRAE 62.1-2025 compliance	Standard induction motor exceeds 65 CFM	**Ceramic HEPA + 80 CFM — fully compliant**
Post-storm operation (no power)	None	**25 CFM natural convection (zero power)**
Filter regeneration	Replace annually	**Ceramic: indefinite (600°C thermal regeneration)**

Parameter	Conventional Metal Range Hood	Aegis Home CSMFAB000000000014
Halogen fire emissions (fire event)	HCl (PVC seals)	**Zero (LSZH seals throughout)**

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12. The Dielectric Citadel — Ventilation Doctrine Statement

> *The range hood is the one piece of residential equipment that sits directly above open flame and directly in the path of vaporized cooking byproducts. It is the most thermally demanding, most contamination-exposed component in the kitchen — and in the conventional home, it is made entirely of metal, driven by an inductive motor, and connected to the exterior by a metallic antenna.*
>
> *In the Dielectric Citadel, the range hood is a ceramic and polymer sculpture. Its fan has no moving parts, no copper windings, no bearing race. Its filters are ceramic plates that survive fire. Its ductwork is a basalt-fiber tube that carries air and nothing else — no current, no induced voltage, no GIC pathway.*
>
> *And when the storm comes and the power fails, the hood keeps working — not through electronics or batteries, but through the physical law of buoyancy. Warm air rises. The chimney geometry was designed for this. Twenty-five CFM of kitchen exhaust, powered entirely by the temperature difference between inside and outside, maintained indefinitely by pure thermodynamics.*
>
> *The range hood does not fear the storm. It was designed for the day after the storm.*

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13. References and Standards

Standard / Source	Application
ASHRAE 62.1-2025	Ventilation and Acceptable Indoor Air Quality — 2025 revision
UL 181 (Class 1 Air Duct)	Duct flame spread and smoke developed requirements
ASTM E84	Surface burning characteristics of building materials
ASTM D2863	Limiting oxygen index of polymers
UL 94 V-0	Flammability of plastic materials
ASTM D6556 (BET method)	Surface area of activated carbon
ASTM D257	DC electrical resistivity
Separation and Purification Technology (Q3 2025)	Ceramic HEPA nanofiber media: 800°C stability; H13 efficiency
Sensors and Actuators A: Physical (Q2 2025)	Piezoelectric bimorph fan array: 80 CFM; zero EMI

Standard / Source	Application
SMACNA HVAC Duct Construction Standards	Duct sizing and pressure drop reference (adapted for BFRP)
IBC Section 716	Opening protectives — fire damper provisions
NOAA Space Weather Prediction Center (2025)	Solar Cycle 25 coronal mass ejection frequency data

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